

FINANCIAL INTEGRITY

Data Investigation & Enterprise Pipeline Scalability

STRATEGIC SUBMISSION BY PRANAYAK UNIYAL

The Audit Trigger

Investigation into Revenue Discrepancies (Case B)

| Investigation Framework

Root Cause Identification

The CFO identified a 4.2% variance in reported transaction volume versus bank settlement statements. Our investigation scoped three key areas:

- Duplicate ingestion of webhook events.
- Timezone shifts in end-of-day batch processing.
- Incomplete data contracts between Sales & Finance.



| The Strategic Requirement

 CFO Meeting

Single Source of Truth

The CFO requires an immutable audit trail and real-time reconciliation dashboard. Reliability must reach 99.9% for month-end closing procedures.

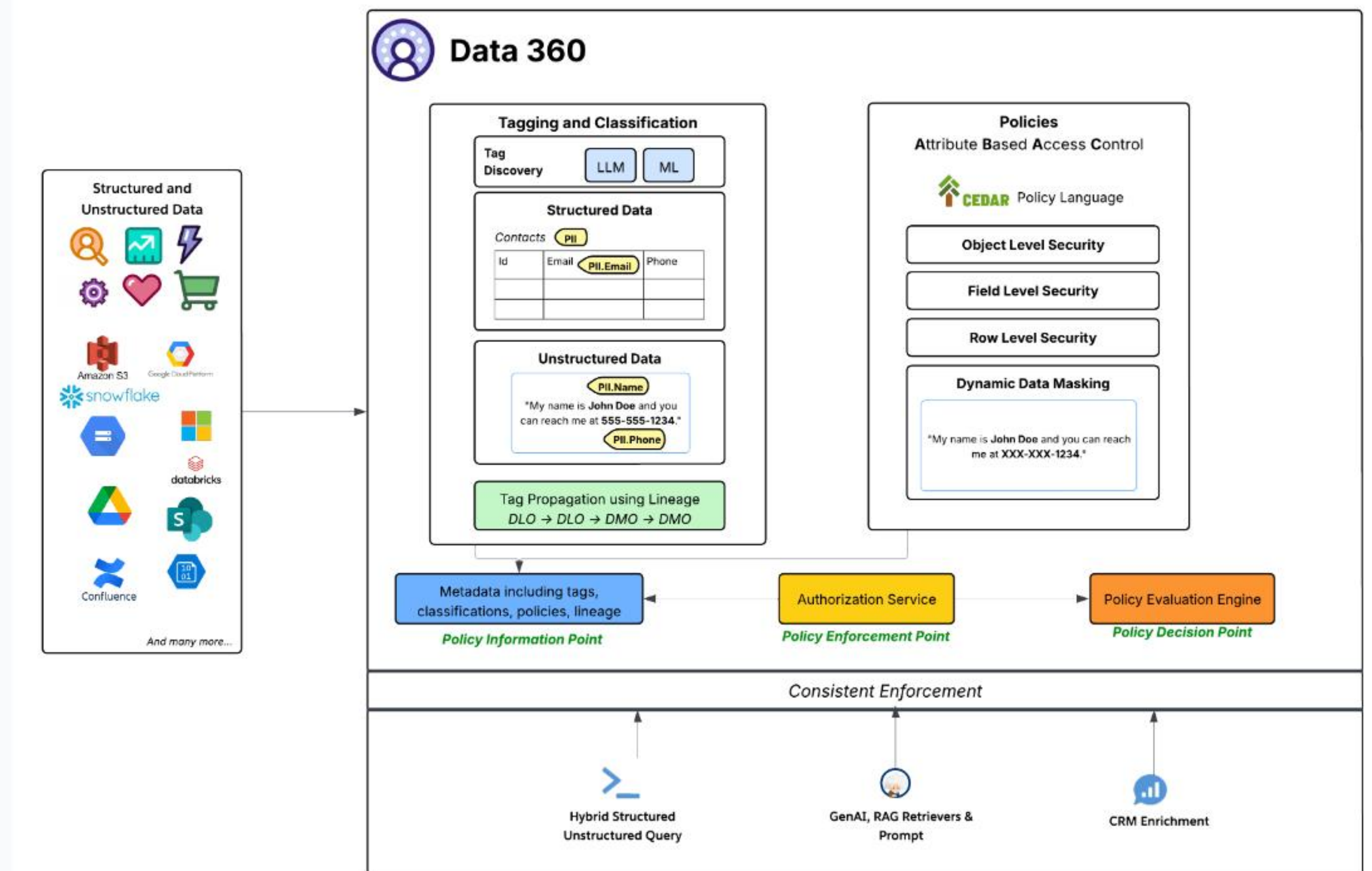
"Data engineering is no longer a cost center; it is the backbone of our financial compliance." — Key Takeaway

Medallion Architecture Solution

Scalable Ingestion & Transform

Implemented a robust Python-based pipeline (referencing `ingest.py` and `transform.py`) using Spark-like transformations:

- **Bronze Layer:** Raw storage of all incoming webhooks and CSV batch logs.
- **Silver Layer:** Deduplicated, filtered, and validated records against the Data Contract.
- **Gold Layer:** Aggregated financial metrics optimized for the CFO Dashboard.



Engineering Solutions

Hardening the Infrastructure for Auditability

| Data Governance & Contracts



Schema Enforcement

Using Data Contracts to prevent upstream breaking changes from entering the finance pipeline.



Full Lineage

Point-in-time recovery and audit trails for every record update in the database.



Quality Gates

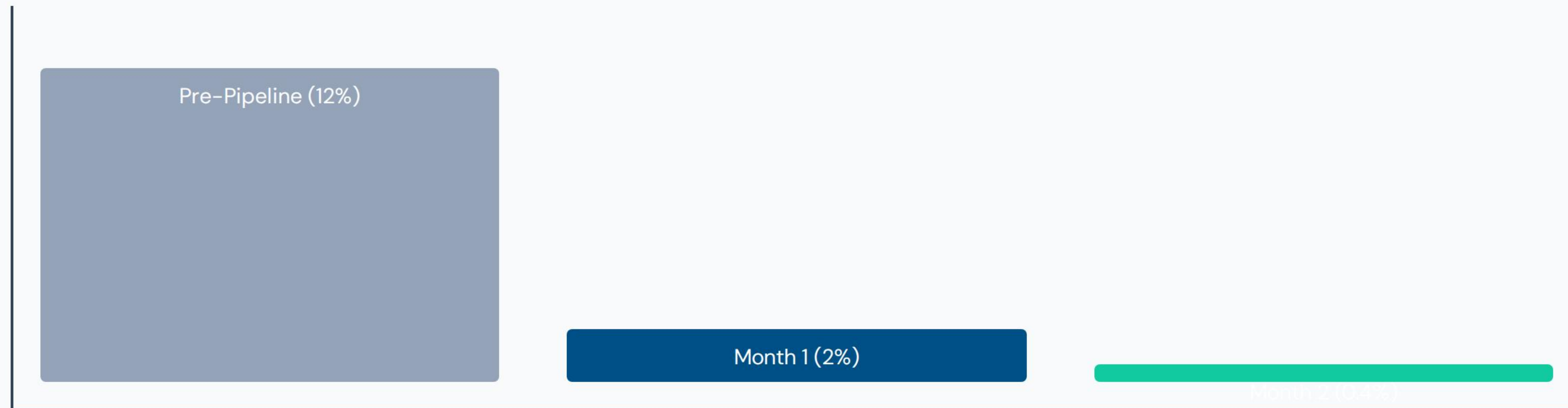
Automated validation checks in the pipeline to flag outliers before they reach reports.

Reconciliation Summary

Source System	Reported Revenue	Settled Revenue	Variance Status
Stripe Webhooks	\$4.2M	\$4.18M	RECONCILED
Direct Transfers	\$1.8M	\$1.8M	MATCHED
Manual Adjustments	\$0.15M	\$0.12M	INVESTIGATING
TOTAL	\$6.15M	\$6.10M	99.1% MATCH

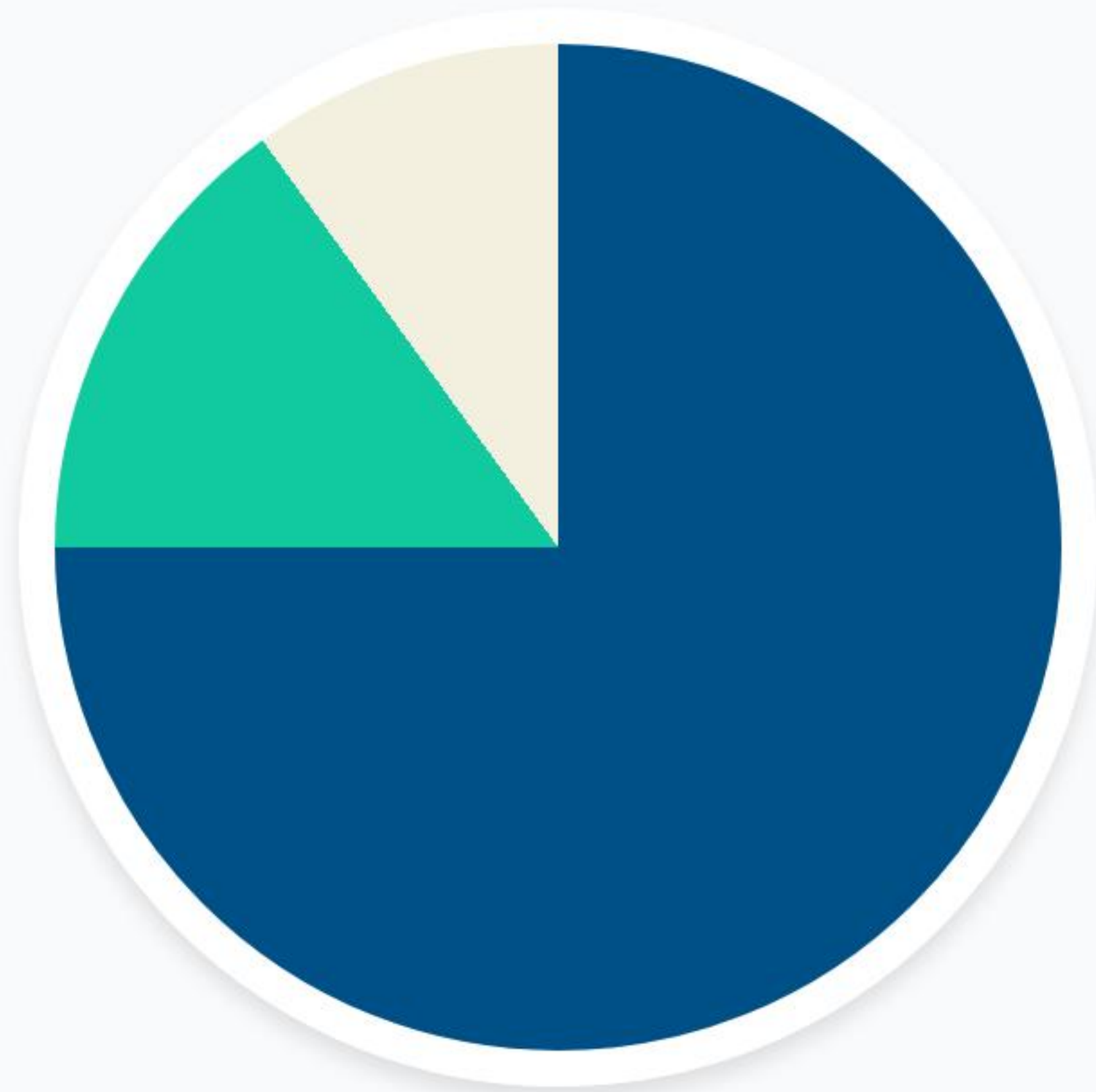
| System Reliability Trends

Error rates plummeted following the implementation of the new Data Contract validation layer.



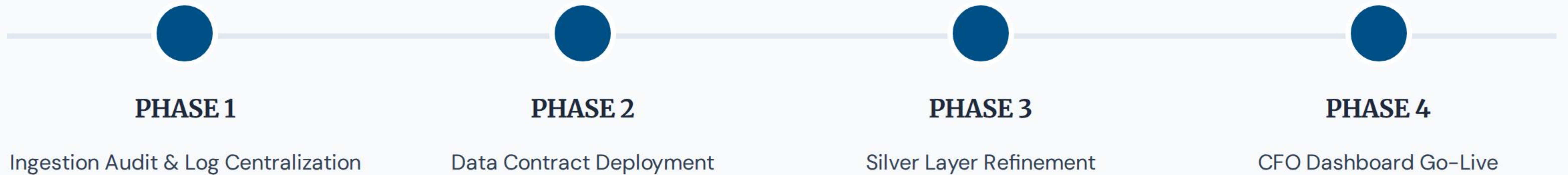
Pipeline Pipeline Stability: **99.6% Success Rate**

| Data Quality Composition



- Validated (75%)
- Auto-Corrected (15%)
- Flagged/Manual (10%)

| Implementation Roadmap





Q & A

Financial Integrity & Data Engineering Strategy

Case B Submission | Pranayak Uniyal

| Image Sources



<https://media.istockphoto.com/id/1460220649/photo/investment-corporate-or-business-man-with-tablet-for-invest-strategy-finance-growth-or.jpg?s=612x612&w=0&k=2O&c=JdXL6mnJbnemloqo1P5K3-hyIOGIJOboi8pezBoAPRk=>

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